Question	Answer	Marks	Guidance
5(a)	$I_T = I_1 + I_2 + I_3$ either $(V / R_T) = (V / R_1) + (V / R_2) + (V / R_3)$ or $(I_T / V) = (I_1 / V) + (I_2 / V) + (I_3 / V)$	**B1**	Throughout allow I and R without subscript to mean I_T and R_T respectively, and allow V_T for V
	<u>and</u> $1 / R_T = 1 / R_1 + 1 / R_2 + 1 / R_3$	**B1**	If uses V_1, V_2, V_3 must also explicitly state that $(V =)\ V_1 = V_2 = V_3$. For the second mark, the associated working <u>and</u> final answer must both be shown A fully correct answer in terms of two rather than three resistors credit 1 / 2
5(b)(i)	$R_{(T)} = (1 / 2.8 + 1 / 2.8)^{-1}$ $= 1.4$ current in battery $= V / R_{(T)}$ $= 12 / 1.4$	**C1**	Allow any correct expression for the parallel combination e.g. $1 / R_{(T)} = 1 / 2.8 + 1 / 2.8$ or $R_{(T)} = (2.8 \times 2.8) / (2.8 + 2.8)$
	= 8.6 A	**A1**	
	OR $I = 12 / 2.8$ $= 4.3$	**(C1)**	
	current in battery = 4.3 + 4.3 = 8.6 A	**(A1)**	<u>Special case:</u> A final answer of 4.3 A without any supporting working credit 1 / 2

Question	Answer	Marks	Guidance
5(b)(ii)	$v = I / Anq$ $= (8.6 / 2) / (6.1 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$	**C1**	no mark for symbol formula as given in the formula sheet ECF current from **5(b)(i)** Allow $(12 / 1.4) / (2 \times 6.1 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$ if treats as a single wire of double area
	$= 5.2 \times 10^{-5}$ m s^{-1}	**A1**	Special case: candidates that do not half their current from **5(b)(i)** can be credited 1 / 2 if working is shown e.g. $= 8.6 / (6.1 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$ $= 1.0 \times 10^{-4}$ m s^{-1}
5(c)(i)	light-dependent resistor	**B1**	Allow LDR
5(c)(ii)	$I_{(X)} = 5 - 12 / 2.8$ $= 0.71$	**C1**	Allow $I_{(X)} = 5 - (8.6 / 2)$ (= 0.70) and allow ECF of current from **5(b)(i)**
	$R_{(\text{top branch})} = 12 / 0.71$ $= 16.8$	**C1**	
	$R_{(X)} = 16.8 - 2.8$ $= 14\ \Omega$	**A1**	
	OR $I_{(X)} = 5 - 12 / 2.8$ $= 0.71$	**(C1)**	Allow $I_{(X)} = 5 - (8.6 / 2)$ (= 0.70) and allow ECF of current from **5(b)(i)**
	$V_{(\text{resistance wire})} = 0.71 \times 2.8$ $= 2.0$ $V_{(X)} = 12 - 2.0$ $=10$	**(C1)**	
	$R_{(X)} = 10 / 0.71$ $= 14\ \Omega$	**(A1)**	

Question	Answer	Marks	Guidance
5(c)(ii)	OR $R_{(T)} = 12 / 5$ $= 2.4$	(C1)	
	$1 / 2.4 = 1 / 2.8 + 1 / R_{(\text{top branch})}$ $R_{(\text{top branch})} = 16.8$	(C1)	Allow $1 / 2.4 = 1 / 2.8 + 1 / (2.8 + R_{(X)})$
	$R_{(X)} = 16.8 – 2.8$ $= 14\ \Omega$	(A1)	
5(c)(iii)	(current in battery decreased, total resistance increased,) resistance (of X) increased	M1	Allow 'LDR' instead of X Allow V, I, R to mean potential difference, current and resistance respectively Allow ↓ for decreases and ↑ for increases (BOD) Ignore all comments about p.d. and emf
	light intensity (incident on X) decreased	A1	Allow '(environment is) darker / dimmer / less light / less bright' Allow 'temperature (of X/LDR) has decreased'